

Hong Kong University of Science and Technology
COMP 4211: Machine Learning
Fall 2025

Programming Assignment 1

Due: 26 September 2025, Friday, 11:59pm

1 Objectives

The objectives of this programming assignment are:

- To practice data importing and preprocessing skills using the `pandas` library.
- To better understand supervised learning methods by using `scikit-learn`.
- To evaluate the performance of several supervised learning methods by conducting an empirical study on a real-world dataset.

2 Dataset

Variable	Description
dteday	Date of observation
season	Season (Winter, Spring, Summer, Fall)
yr	Year (0: 2011, 1: 2012)
mnth	Month (1–12)
hr	Hour of the day (0–23)
holiday	Whether day is a holiday (weekends are not counted as holiday)
weekday	Day of the week
workingday	1 if not holiday and is weekday, else 0
weathersit	Weather situation (Clear, Mist/Cloudy, Light Snow/Rain, Heavy Rain/Snow)
temp	Temperature (°C)
atemp	Feeling temperature (°F)
hum	Humidity (%)
windspeed	Wind speed
uv	UV index
cnt	Total rental bikes

Table 1: Variable descriptions

You will use an anonymous bike sharing dataset provided as a ZIP file (data.zip). There are two CSV data files: `train.csv` (training set) and `test.csv` (test set). The ‘`cnt`’ column of the CSV files is the **regression target**, while the ‘`uv`’ column is the **classification target** with 13 labels: 0, 1, $\dots$, 12.

3 Major Tasks

The assignment consists of four parts (plus an optional bonus part) and a written report:

PART 1: Use `pandas`, `scipy`, `matplotlib`, or `seaborn` for data importing and exploration.

PART 2: Use `scikit-learn` for data preprocessing.

PART 3: Use the linear regression model and feedforward neural networks for regression.

PART 4: Use the logistic regression model and feedforward neural networks for classification.

BONUS: Do additional feature engineering and use `scikit-learn` to tune the hyperparameters and validate the data preprocessing.

WRITTEN REPORT: Report the results and answer some questions in the Jupyter Notebook.

More details will be provided in the following sections. Note that **Q n** refers to a specific question (the n th question) that you need to answer in the written report. **B n** refers to the bonus questions that you may answer to get extra marks. If you lose points in Parts 1 to 4, the bonus points can help recover your score up to the maximum (100). Any marks earned beyond 100 will not be counted. All the coding work should be done using Python 3.

4 Part 1: Data Exploration and Preparation

Q1. Load and Merge Data. You are given four files: `train.csv`, `train_p1.txt`, `train_p2.json`, and `train_p3.csv`, each containing different features. Your task is to load these files and merge `train_p1.txt`, `train_p2.json`, and `train_p3.csv` into a single `pandas` DataFrame by aligning records with the same date and hour. Make sure to handle any duplicate entries in `train_p2.json` appropriately. A merged version (`train.csv`) is provided for sanity checking your result.

Q2. Missing Values.

- **Identification:** Report any features with missing values and the proportion of missing values for them.
- **Potential Impact:** Briefly discuss how these missing values might affect analysis and model performance.

Q3. Feature Distribution.

- **Numerical Features:** Distinguish between discrete and continuous variables. For all numerical features, report their statistical summaries (mean, median, range, and standard deviation) and illustrate their distributions using `seaborn.boxplot` with all default arguments.
- **Categorical Features:** Identify which ones are binary, nominal, or ordinal. Select 2 of the categorical features, summarize the count of categories for each, and visualize their distributions using bar plots.

Q4. Outliers.

- **Detection:** Based on the box plot from **Q3**, identify which features contain potential outliers. For each, show the entries flagged as outliers using a detection method such as Z-score or IQR.

- **Consideration:** Using the box plot of features and/or the detected points, briefly discuss whether these are data errors or genuine extreme conditions, and how they should be treated in preprocessing.

B1. Grubbs' Test / Hypothesis Test. Select one feature with an extreme minimum value. Test whether this minimum value is statistically inconsistent with the rest of the sample, assuming the data (apart from at most one outlier) are approximately normal. Use $\alpha = 0.05$.

$$H_0 : \text{All } x_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2). \quad H_1 : x_{\min} \text{ is an outlier (left-tail).}$$

1. Compute the sample mean $\bar{x}$ and unbiased standard deviation s using *all* n observations.
2. Form the (one-sided, left-tail) Grubbs statistic is defined as:

$$G = \frac{\bar{x} - x_{\min}}{s}.$$

3. Reject H_0 (flag the minimum as an outlier) if

$$G > G_{\alpha, n}^{(\text{one-sided})} = \frac{n-1}{\sqrt{n}} \sqrt{\frac{t_{\alpha/n, n-2}^2}{n-2 + t_{\alpha/n, n-2}^2}},$$

where $t_{\alpha/n, n-2}$ is the upper α/n quantile of t distribution with $n-2$ degrees of freedom.

4. Report the value of G and $G_{\alpha, n}$ and your conclusion.

Q5. Correlation Analysis.

- **Feature Correlation:** Visualize the correlation between every two features (numerical features only, including the regression targets) with a heatmap. You can use `matplotlib` and `seaborn` to help you.
- **Insights:** Highlight any strong correlations that might influence feature selection or necessitate feature engineering.

5 Part 2: Data Preprocessing Techniques

Preprocessing is an indispensable step in the machine learning pipeline, serving as the foundation upon which the performance of your models is built. It encompasses a range of techniques to transform raw data into a more suitable and effective format for modeling. Proper preprocessing can significantly enhance model accuracy, efficiency, and interpretability by addressing missing values, inconsistent data formats, and irrelevant or redundant features.

In this task, you will apply a variety of advanced preprocessing techniques from `scikit-learn` to your dataset. You are expected to understand the rationale behind each technique and its impact on the data to predict the effect on subsequent model performance. In addition to the required APIs, you can additionally use and report on any other preprocessing and normalization modules in `scikit-learn`. You will have a chance to test these conjectures in the optional bonus part.

Q6. Handling Missing Values. Apply `SimpleImputer` to handle missing values. Briefly discuss its impact on feature distribution and model performance. Briefly discuss what imputation strategies (mean, median, mode, and constant) should be used for what reasons.

Q7. Normalization and Standardization. Normalize/standardize different types of numerical features using `StandardScaler`, `MinMaxScaler`, or `RobustScaler`. Briefly discuss the difference between these techniques and outline scenarios where each is most suitable. Finally, explain why it is generally better to normalize/standardize features rather than working directly with raw values.

Q8. Encoding Categorical Variables. Utilize `OneHotEncoder` and `OrdinalEncoder` to encode different types of categorical variables. For certain ordinal feature(s), a direct application of the `OrdinalEncoder` may lead to a suboptimal ranking between categories. You need to handle this by adjusting the settings within the `OrdinalEncoder`. Additionally, you need to report the entire table in the jupyter notebook, showing the first 10 samples before and after processing in `pandas.DataFrame` structure. Briefly explain the scenarios where each encoding technique is preferred.

Q9. Handling Duplicate Features. Check if there are duplicate or features that contain redundant information in the dataset. Keep, drop, or consolidate them, and provide a justification for your decision. Report the features you identified as duplicates or highly correlated, and explain the potential impact of keeping them versus removing them.

Q10. Feature Selection: Perform feature selection (twice, once for each target) with either feature selection modules in `scikit-learn`, such as `SelectKBest`, or based on the correlation analysis on all features (including numerical and categorical features) similar to **Q5**. Describe how feature selection impacts the performance of your models. You need to report which feature(s) were removed for what reasons.

6 Part 3: Regression

6.1 Linear Regression

In this task, you will build multiple linear regression models in the first step, where each model uses only one feature to determine its correlation with the regression target (`'cnt'`). In this stage, you only need to focus on binary categorical and continuous numerical features.

Then, in the second step, you will build another linear regression model using the features that yield the best performance in your experiment to explore their relationship with the regression target.

You are required to use the `train_test_split` submodule in `scikit-learn` to split the data in `train.csv`, with 80% for training and 20% for validation. You should set `random_state = 4211` for reproducibility.

Q11. After training multiple models described above using the training set, use them to make predictions on the validation set. Report the validation R^2 score and Mean Squared Error (MSE) for each of the models to evaluate the relationship between different features and the regression target. Explain the relationship and difference between R^2 score and Mean Squared Error (MSE).

Q12. Discuss the mathematical meaning of the weight (in brief) for a binary categorical independent variable. How can a nominal categorical feature with more than 2 possible values be formulated as the independent variable of a linear model? Include the necessary component of the nominal categorical features alongside the previously selected features to enhance the model's performance compared to **Q11**. Report the validation R^2 score and MSE of this linear regression model.

B2. Interpreting a regression coefficient via partialling out. Consider the variables `cnt` (response), `temp`, and `hum` (covariates). All regressions below include an intercept.

1. Fit the multiple linear regression

$$\widehat{\text{cnt}} = \hat{\beta}_0 + \hat{\beta}_{\text{temp}}\text{temp} + \hat{\beta}_{\text{hum}}\text{hum}$$

and record the estimated $\hat{\beta}_{\text{hum}}$.

2. Fit the simple regression

$$\widehat{\text{cnt}} = \hat{\alpha}_0 + \hat{\alpha}_{\text{temp}}\text{temp}$$

and define the residual vector

$$\mathbf{r}_{\text{cnt}|\text{temp}} = \text{cnt} - \widehat{\text{cnt}}.$$

3. Fit the simple regression

$$\widehat{\text{hum}} = \hat{\gamma}_0 + \hat{\gamma}_{\text{temp}}\text{temp}$$

and define the residual vector

$$\mathbf{r}_{\text{hum}|\text{temp}} = \text{hum} - \widehat{\text{hum}}.$$

4. Finally, regress the residuals from step (2) on the residuals from step (3):

$$\widehat{\mathbf{r}}_{\text{cnt}|\text{temp}} = \hat{\delta}_0 + \hat{\delta}_1 \mathbf{r}_{\text{hum}|\text{temp}}$$

What do you observe about the estimates $\hat{\delta}_0$ and $\hat{\delta}_1$ relative to $\hat{\beta}_{\text{hum}}$ from step (1)? Explain your findings.

6.2 Feedforward Neural Networks

In this part, you are asked to use the features selected in **Q10** to train feedforward neural networks.

You need to build different neural networks with three hidden layers, each having the same number of hidden units H , where $H \in \{1, 8, 32, 128\}$. The hyperparameter `early_stopping` can be set to 'True' to avoid overfitting (default is 'False'). The other hyperparameters may just take their default values. For each hidden layer in a specific neural network model, the number of hidden units should be kept the same for simplicity. After training, evaluate your models by reporting the R^2 scores on the validation set.

Q13. For each value of H , report the neural network model configuration and its corresponding performance. Each model should be trained once, and you should present the R^2 and MSE results and plot them against H using `matplotlib` on the validation set.

Q14. Do you notice any trend of R^2 and MSE score when you increase the hidden layer size from 1 to 128? Explain the difference between the two metrics in evaluating regression models.

Q15. Try to change the hyperparameter `activation` to be `'identity'`, do you notice any difference in the performance of the model? Explain why.

7 Part 4: Classification

In this task, you will build a logistic regression model as well as neural network classifiers to predict the class to which the classification target `uv` belongs. You are also required to use the `train_test_split` submodule in `scikit-learn` to split the data, with 80% for training and 20% for validation. As before, we ask that you set `random_state = 4211` for reproducibility.

7.1 Logistic Regression

Learning of the logistic regression model should use a gradient-descent algorithm by minimizing the cross-entropy loss. It requires that the step size parameter η be specified. Try out a few values ($\eta < 1$) and choose one that leads to stable convergence. You may also decrease η gradually during the learning process to enhance convergence. When set properly, this can be done automatically in `scikit-learn`. Use the features selected in **Q10** to train the model. During training, record the training time for the logistic regression model. After training, you are required to evaluate your model using accuracy and the F1 score on the validation set. Please be careful to choose the built-in models that are suitable for your tasks, i.e., the default setting of `sklearn.linear_model.LogisticRegression` is *not* a correct choice for our logistic regression model since it does not use gradient descent.

Q16. Report the model setting, training time, and performance of the logistic regression model. Since the solution found may depend on the initial weight values, you are expected to try three different parameter combinations (e.g. `penalty` change to `l1`, set `warm_start = True`, ...) except the `loss` needs to be fixed as `log_loss` and `random_state = 4211` and report the corresponding mean and standard deviation of the training time, accuracy, and the F1 score for each setting.

Q17. Plot the ROC curve for each class calculated on the validation set with the last model in [Q16] and report the AUC value.

Q18. Plot the confusion matrix on the validation set with the last model in [Q16]. Discuss why these numbers are helpful in comparison to solely relying on the F1 score.

7.2 Feedforward Neural Networks

Neural network classifiers generalize logistic regression by introducing one or more hidden layers. Their learning algorithm is similar to that for logistic regression, as described above.

You need to build different neural networks with three hidden layers, each having the same number of hidden units H , where $H \in \{1, 8, 32, 128\}$. The hyperparameter `early_stopping` can be set to `'True'` to avoid overfitting (default is `'False'`). The other hyperparameters may just take their default values. After training, evaluate your models using accuracy and the F1 score on the validation set.

Q19. Use the best model trained above to calculate the values of the micro precision and

recall on the validation set. Explain why the F1 score is calculated by the harmonic mean of precision and recall $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$ instead of their average $\frac{Precision + Recall}{2}$.

Q20. Report the model setting, accuracy, and the F1 score (both micro and macro) of the neural networks for each value of H .

Q21. Plot the accuracy and F1 score (both macro and micro average) for each value of H . Compare the difference between macro and micro averages of F1 score. Suggest the scenarios when to use each. Suggest a possible reason for the gap between the accuracy and macro F1 score.

Q22. Do you notice any trend when you increase the hidden layer size from 1 to 128? If so, please describe what the trend is and suggest a reason for your observation.

Q23. Plot the ROC curve calculated on the validation set with the hidden layer size of 1 and the hidden layer size of 128. Give one reason why we need to examine the ROC curve as well.

8 Performance Enhancement

In this part, we focus on the enhancement of the classification model (`uv` as the target).

B3. More on Feature Engineering. You are encouraged to explore dimensionality reduction techniques (e.g., principal component analysis) and other feature engineering methods (e.g., interaction terms, polynomial features, binning, or cyclical encoding for features like `hr`, `day`, `mnth`). You can choose to undo the feature selection in **Q10**.

Use the logistic regression classifier in section 7.1, perform an ablation study and discuss how your feature selection and engineering choices affect model performance and interpretability. Clearly report which features were removed, transformed, or added, and explain why.

B4. Hyperparameter Tuning. Use `scikit-learn`'s `GridSearchCV` and `RandomizedSearchCV` with cross-validation to tune the hyperparameters of `MLPClassifier`. Use the features obtained in **B3**.

This is done by randomly sampling 80% of the training instances to train a classifier and then validate it on the remaining 20%. Five such random data splits are performed and the average over these five trials is used to estimate the generalization performance. You are expected to try at least 10 hyperparameter configurations (with each search method having 5). Set the `random_state` hyperparameter of the neural network model to 4211 for reproducibility and `early_stopping` to 'True' to avoid overfitting.

1. Compare the time taken by each method.
2. Report the best validation accuracy with the corresponding hyperparameter configuration achieved by each method.
3. Discuss why one method might perform better in terms of efficiency or accuracy.

B5. Model Selection. Reuse the logistic regression classifier in **B3**. Compare its validation accuracy and F1-score against the best MLP model in **B4**. Give a possible explanation of why the logistic regression performs better / worse than the MLP model (depends on your

result). You may also add the performance of `sklearn.linear_model.LogisticRegression` for comparison.

B6. Prediction. Use the best-performing model to predict `uv` on the test dataset. Output the prediction result as `b6_results.csv`. Marks will be awarded only if the prediction accuracy reaches a certain threshold.

9 Report Writing

Answer [Q1] to [Q23] (additionally [B1] to [B6] if you do the bonus part as well) in the Jupyter Notebook.

10 Some Programming Tips

As is always the case, good programming practices should be applied when coding your program. Below are some common ones but they are by no means complete:

- Using functions to structure your code clearly
- Using meaningful variable and function names to improve readability
- Using consistent styles
- Including concise but informative comments

You are recommended to take full advantage of the built-in classes of `scikit-learn` to keep your code both short and efficient. Proper use of implementation tricks often leads to speedup by orders of magnitude. Also, please be careful to choose the built-in models that are suitable for your tasks, e.g., `sklearn.linear_model.LogisticRegression` is *not* a correct choice for our logistic regression model since it does not use gradient descent.

11 Assignment Submission

Assignment submission should only be done electronically on the Canvas course site.

You should submit all the codes and written answers in a single Jupyter Notebook `pa1.ipynb`. If you tried **B6**, also submit `b6_result.csv`. When multiple versions with the same filename are submitted, only the latest version according to the timestamp will be used for grading. Files not adhering to the naming convention above will be ignored.

12 Grading Scheme

This programming assignment will be counted towards 10% of your final course grade. Note that the plus sign (+) in the last column of the table below indicates that reporting without providing the corresponding code will get zero point. The maximum scores for different tasks are shown below:

Grading scheme	Code	Report
Part 1	9	7
- Load Data (CSV, TXT, JSON) [Q1]	1	
- Handle duplicate entries [Q1]	1	
- Merge the entries by the date and hour [Q1]	1	
- Report features with missing values and their proportion respectively [Q2]	1	
- Report the feature(s) with missing values and discuss its influence [Q2]	1	+1
- Distinguish between discrete and continuous variables [Q3]		1
- Report the distribution of all numerical features [Q3]	1	
- Visualize the distribution of all numerical feature using a box plot [Q3]	1	
- Identify binary, nominal, and ordinal features [Q3]		1
- Summarize the count of each category in 2 of the categorical features and visualize using bar plots [Q3]	1	
- Show the entries flagged as outliers [Q4]		+1
- Discuss whether they are data errors or genuine extreme conditions [Q4]		1
- Discuss how they should be treated in preprocessing [Q4]		1
- Visualize the correlation between every two features with heatmap [Q5]	1	
- Highlight strong correlations [Q5]		+1
Part 2	8	11
- Perform missing value imputation with imputation strategies [Q6]	1	
- Discuss the impact on feature distribution and model performance [Q6]		1
- Briefly judge what imputation should be used and when [Q6]		1
- Perform normalization/standardization using three scalars [Q7]	2	
- Discuss the difference between these techniques and when to use [Q7]		1
- Discuss why it is generally better to normalize/standardize features [Q7]		1
- Encode categorical features using the two encoders [Q8]	2	
- Report the entire table before and after processing [Q8]		+1
- Explain the scenarios where each encoding technique is preferred [Q8]		1
- Keep, drop or consolidate duplicate columns and provide justification [Q9]	1	+1
- Report the duplicate features and explain the potential impact of keeping them versus removing them [Q9]		2
- Perform feature selection [Q10]	2	
- Explain how feature selection impacts the performance [Q10]		1
- Report which features were removed for what reasons [Q10]		1
Part 3	15	15
- Do train test split for the regression task	1	
- Build the linear regression models [Q11]	4	
- Compute the R^2 scores and MSE scores of the linear regression models and explain [Q11]	2	2
- Explain the relationship and difference between R^2 score and MSE [Q11]		2
- Discuss the meaning of the weight for a binary variable [Q12]		2
- Discuss how to formulate the nominal features as the variable of a linear model [Q12]		1
- Enhance the model by including nominal features [Q12]	1	
- Build the feedforward neural network models	4	
- Plot the R^2 and MSE score for each value of H in the feedforward neural network model [Q13]	2	2
- Report the trend of R^2 and MSE score for each value of H [Q14]		2
- Discuss the difference between R^2 and MSE score in evaluation [Q14]		2
- Change activation to be identity , report and explain the difference [Q15]	1	+2

Grading scheme	Code	Report
Part 4	17	18
- Build the logistic regression model by adopting the gradient descent optimization algorithm	4	
- Compute the training time, accuracy, and F1 score of the logistic regression model and report the statistics [Q16]	2	+3
- Report the ROC curve and AUC value on the validation set [Q17]	1	+2
- Plot confusion matrix [Q18]	1	
- Discuss why it is helpful.[Q18]		2
- Build the feedforward neural network model	4	
- Calculate micro precision and recall [Q19]	2	
- Explain why the F1 score is calculated by the harmonic mean [Q19]		2
- Compute the accuracy and F1 score for each value of H in the feedforward neural network model [Q20]	1	+2
- Plot the accuracy and F1 score with different values of H for the feedforward neural network model [Q21]	2	
- Suggest the scenarios when to use each [Q21]		2
- Explain the gap [Q21]		1
- Performance trend across different H and reason behind [Q22]		+2
- Compare ROC curves and discuss benefit of examining it [Q23]		+2
Bonus Questions	6	9
- Grubb's test [B1]	2	+1
- Fit all the models and compute the corresponding residual vectors [B2]	1	
- Explain the reason of the observation [B2]		+2
- Explore more on feature engineering, then report and explain [B3]		+3
- Compute the time for each method [B4]	2	
- Report the best accuracy with its configuration [B4]	1	
- Explain the results [B4]		1
- Compare the logistic regression models, then explain [B5]		1
- Use the best model to classify the test set [B6]		1

Late submission will be accepted but with penalty.

The late penalty is deduction of one point (out of a maximum of 100 points) for every hour late after 11:59pm with no more than two days (48 hours). Being late for a fraction of an hour is considered a full hour. For example, two points will be deducted if the submission time is 01:23:34.

13 Academic Integrity

Please refer to the regulations for student conduct and academic integrity on this webpage: <https://registry.hkust.edu.hk/resource-library/academic-standards>.

While you may discuss with your classmates on general ideas about the assignment, your submission should be based on your own independent effort. In case you seek help from any person or reference source, you should state it clearly in your submission. Failure to do so is considered plagiarism which will lead to appropriate disciplinary actions.